

Update for Mr. Ternes

What's the research in the Milner Group about?

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Outline

- 1 Introduction
- 2 The optical centrifuge
- 3 Droplets and Molecules
- 4 Measurement Technique
- 5 Some Data

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A view key ideas

- **What are we trying to do?**

- Spinning molecules in superfluid ^4He
- Use different methods to bring energy in this microscopic system
- compare with free molecules in the gas phase

- **Goal of the research**

- figure out properties of superfluidity, such as: friction, interaction with molecules, breakdown of superfluidity

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Animation of an accelerating optical centrifuge

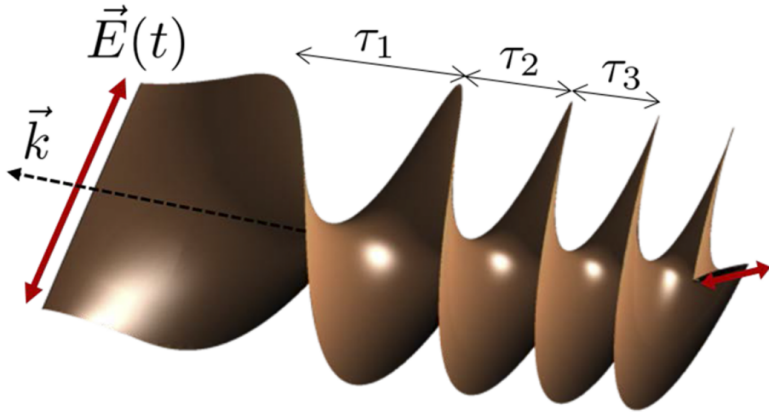


Figure: E-field of an accelerating optical centrifuge¹

¹MacPhail-Bartley et al. 2020.

How to make an optical centrifuge

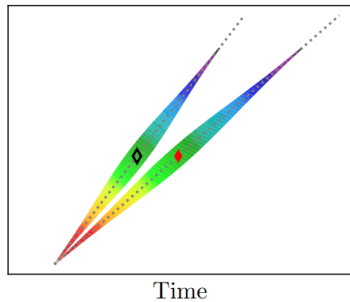


Figure: Chirped pulses used to create the accelerating rotating electric field^a

- Take two pulses with **equal amplitudes** and **opposite circular polarizations**: one right-circular (RCP), one left-circular (LCP).
- At any instant, **RCP + LCP of equal strength** adds up to a **linearly polarized** electric field.
- The **only thing that matters** for the *direction* of that linear polarization is the **relative optical phase** between the two components.
- If that relative phase changes in time, the resulting linear polarization axis **rotates in time**.

^aWang et al. 2025.

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^4He nanodroplets & field coupling to linear molecules

- **^4He nanodroplets:** formed by supersonic expansion through a cryogenic nozzle; cool to $\sim 0.4\text{ K}$ and are typically **superfluid**.
- **Size:** usually $N \sim 10^3\text{--}10^6$ **He atoms** $\Rightarrow$ radius $10 - 100\text{ nm}$ (source-condition dependent)².
- **Molecule pickup:** droplets capture molecules in a pickup cell and rapidly **cool/thermalize** them.
- **E-field coupling (nonresonant):** alignment via anisotropic polarizability³

$$U(t) \propto -\Delta\alpha E^2(t) \cos^2 \theta, \quad \Delta\alpha = \alpha_{\parallel} - \alpha_{\perp}.$$

- For $\Delta\alpha > 0$ the molecule aligns with the instantaneous polarization axis; if that axis rotates (optical centrifuge), the molecule can be **captured and spun up** to high J .

²Toennies and Vilesov 2004.

³MacPhail-Bartley 2024.

Spinning of NO dimers

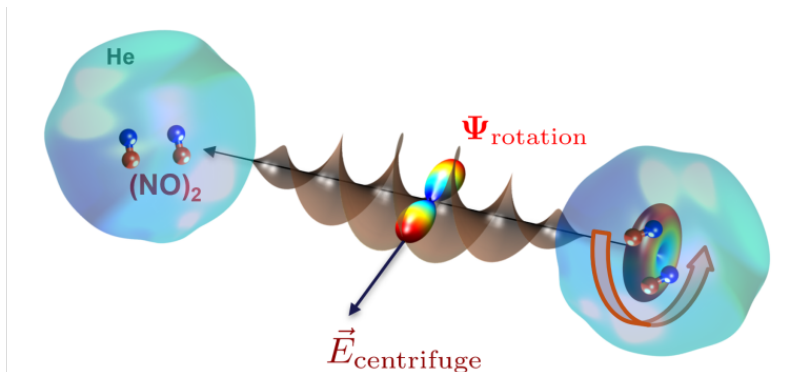


Figure: Spinning of NO dimers in helium nanodroplets with optical centrifuge⁴

⁴Home Page | Ultrafast Coherent Control Group 2026.

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Measurement principle: Coulomb explosion + VMI readout

- **Prepare:** Molecules are embedded in ^4He nanodroplets and are aligned/spun up by the pump (e.g. optical centrifuge) at a controlled pump–probe delay.
- **Coulomb explosion (probe):** A short, intense probe pulse removes multiple electrons $\Rightarrow$ the molecule becomes a **highly charged ion** and fragments.
- **Axial recoil:** Fragment ions recoil mainly along the **molecular bond axis** (for suitable channels) $\Rightarrow$ ion emission angles encode the instantaneous molecular orientation.
- **Velocity Map Imaging (VMI):** Electrostatic lenses map the 3D ion velocity distribution onto a 2D detector: ions with the same transverse velocity hit the same detector radius.
- **Detection & analysis:** The detector image (MCP/phosphor + camera) gives **angular distributions** (alignment/orientation) and **kinetic energy release**; repeating vs delay reconstructs the rotational dynamics.

Example of one measurement frame

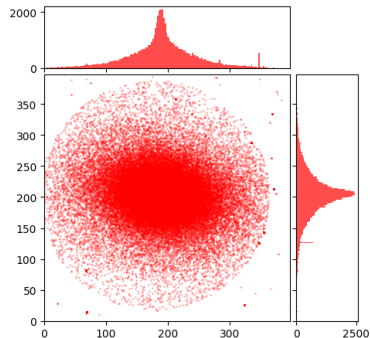


Figure: Detected ions (unprocessed)

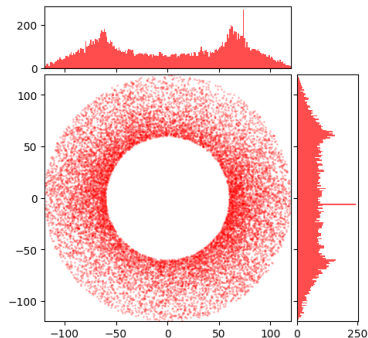


Figure: Detected ions (processed)

- We scan the pump-probe delay

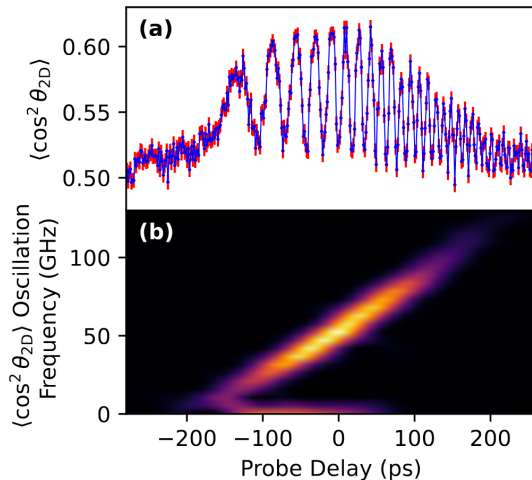
$$\tau = t_{\text{probe}} - t_{\text{pump}}$$

to map the time-dependent rotational dynamics.

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Measurement data from a typical scan



- $\cos^2(\Theta)$ data from a typical scan
- centrifuge accelerates from 0 **GHz** to about 120 **GHz**
- central frequency of the pump pulse approx. 50 **GHz**

Questions?

Thank you for your attention!

-  [Home Page | Ultrafast Coherent Control Group](https://coherentcontrol.sites.olt.ubc.ca) (Jan. 2026). [Online; accessed 7. Jan. 2026]. URL: <https://coherentcontrol.sites.olt.ubc.ca>.
-  MacPhail-Bartley, Ian (June 2024). “Control of Molecular Rotation in Superfluid Helium”. PhD thesis. Vancouver, BC, Canada: The University of British Columbia.
-  MacPhail-Bartley, Ian et al. (Apr. 2020). “Laser control of molecular rotation: Expanding the utility of an optical centrifuge”. In: *Review of Scientific Instruments* 91.4. ISSN: 1089-7623. DOI: 10.1063/1.5140358. URL: <http://dx.doi.org/10.1063/1.5140358>.
-  Toennies, J Peter and Andrey F Vilesov (2004). “Superfluid helium droplets: a uniquely cold nanomatrix for molecules and molecular complexes”. In: *Angewandte Chemie International Edition* 43.20, pp. 2622–2648.
-  Wang, Kevin et al. (2025). *An ultraslow optical centrifuge with arbitrarily low rotational acceleration*. arXiv: 2512.20568 [physics.chem-ph]. URL: <https://arxiv.org/abs/2512.20568>.